

Assignment
Data management & visual insight
Sem - 5

Assignment Instructions

1. All students **must complete all the sums** with full steps and clear explanations.
2. Your work should be **neat and clean** — present every solution properly.
3. After completing the assignment, **create a PDF** containing all the solved sums.
4. Submit the PDF to the **given Google submission link**.
5. **Last date for submission: 25th November**
Assignments submitted on or before the due date will be accepted.
Assignments submitted after the due date **will not be accepted**.

Question

1. A company produces rechargeable batteries. A random sample of **36 batteries** is tested, and the **average lifetime** is found to be **82 hours**. The **population standard deviation (σ)** is known to be **6 hours**.

Find:

- The **point estimate** of the population mean.
 - The **standard error** of the mean.
 - The **margin of error** for a 95% confidence level.
 - The **95% confidence interval** for the population mean battery life.
2. A survey of **64 households** reveals that the **average monthly electricity bill** is **\$120**. The **population standard deviation** is **\$20**.

Find:

- Point estimate of the population mean monthly bill.
 - Standard error of the mean.
 - Margin of error for a **90% confidence level**.
 - Construct the **90% confidence interval** for the true mean monthly electricity bill.
3. An online store wants to estimate its average delivery time. A sample of **49 orders** has an **average delivery time** of **3.4 days**, with a known **population standard deviation of 0.7 days**.

Find:

- Point estimate of the population mean delivery time.
 - Standard error of the mean.
 - Margin of error for a **99% confidence level**.
 - 99% confidence interval for the mean delivery time.
4. A professor wants to estimate the mean score of her class on the final exam. A sample of **25 students** has a mean score of **76**. The **population standard deviation** is known to be **8**.

Find:

- Point estimate of the population mean exam score.

- Standard error of the mean.
- Margin of error for a **95% confidence level**.
- Construct the **95% confidence interval** for the true mean exam score.

5. A coffee shop owner wants to estimate how much coffee customers drink on average per day. A random sample of **100 customers** shows an average consumption of **2.8 cups per day**. The **population standard deviation** is **0.5 cups**.

Find:

- Point estimate of the population mean daily coffee consumption.
- Standard error of the mean.
- Margin of error for a **99% confidence level**.
- The **99% confidence interval** for the true mean daily coffee consumption.

6. A university wants to estimate how many hours students study per week. A sample of **16 students** gives a **mean of 24 hours** with a **sample standard deviation (s)** of **4.5 hours**.

Find:

- Point estimate of the population mean study hours.
- Standard error of the mean.
- Margin of error for a **95% confidence level** (use *t*-distribution).
- The **95% confidence interval** for the population mean study hours.

7. A quality control inspector weighs **10 apples** randomly selected from a shipment. The **average weight** is **150 grams** with a **sample standard deviation** of **8 grams**. **Find:**

- Point estimate of the population mean apple weight.
- Standard error of the mean.
- Margin of error for a **99% confidence level**.
- Construct the **99% confidence interval** for the mean apple weight.

8. A sample of **25 employees** from a large firm reports a **mean daily commute time** of **42 minutes** with a **sample standard deviation** of **7 minutes**.

Find:

- Point estimate of the population mean commute time.
- Standard error of the mean.
- Margin of error for a **90% confidence level**.
- 90% confidence interval for the population mean commute time.

9. A bank wants to estimate the average waiting time for its customers. A sample of **20 customers** shows a **mean waiting time** of **5.8 minutes** and a **sample standard deviation** of **1.2 minutes**.

Find:

- Point estimate of the population mean waiting time.
- Standard error of the mean.
- Margin of error for a **95% confidence level**.

- Construct the **95% confidence interval** for the true mean waiting time.

10. A sample of **30 students** from a college course has an **average exam score** of **78** and a **sample standard deviation** of **10**.

Find:

- Point estimate of the population mean exam score.
- Standard error of the mean.
- Margin of error for a **99% confidence level**.
- 99% confidence interval for the population mean exam score.

11. A company surveys **200 customers** about their new product. **154 customers** say they are satisfied with it.

Find:

- The **point estimate** of the population proportion of satisfied customers.
- The **standard error** of the sample proportion.
- The **margin of error** for a **95% confidence level**.
- The **95% confidence interval** for the true proportion of satisfied customers.

12. In a poll of **1,000 registered voters**, **620** say they support a new policy. **Find:**

- Point estimate of the proportion of voters supporting the policy.
- Standard error of the proportion.
- Margin of error for a **90% confidence level**.
- Construct the **90% confidence interval** for the true proportion of support.

13. A factory inspects a sample of **400 bulbs** and finds **28 defective** ones. **Find:**

- Point estimate of the proportion of defective bulbs.
- Standard error of the sample proportion.
- Margin of error for a **99% confidence level**.
- 99% confidence interval for the population proportion of defective bulbs.

14. Out of **500 applicants**, **315** were accepted into a college program.

Find:

- Point estimate of the admission proportion.
- Standard error of the proportion.
- Margin of error for a **95% confidence level**.
- Construct the **95% confidence interval** for the true admission proportion.

15. A survey of **150 households** shows that **132** have access to the internet. **Find:**

- Point estimate of the proportion of households with internet access.
- Standard error of the sample proportion.
- Margin of error for a **99% confidence level**.
- The **99% confidence interval** for the population proportion of households with

internet access.

16. A manufacturer claims that the **average lifetime** of its light bulbs is **1,200 hours**. A random sample of **36 bulbs** has a **mean lifetime of 1,165 hours**, and the **population standard deviation** is **80 hours**.
At the **5% significance level**, test whether the **mean lifetime differs from 1,200 hours**.
17. A company claims that their new battery lasts **at least 10 hours** on average. A random sample of **49 batteries** has a **mean of 9.6 hours** and a **population standard deviation of 1.2 hours**.
At $\alpha = 0.05$, test whether the **mean battery life is less than 10 hours**.
18. The average score on a national exam is **75**. A teacher believes her students score **lower than average**.
A sample of **20 students** shows a **mean score of 72** with a **sample standard deviation of 6**.
At $\alpha = 0.05$, test the teacher's claim.
19. A nutritionist claims that the average daily sodium intake for adults is **2,300 mg**. A sample of **25 adults** shows a **mean intake of 2,420 mg** and a **sample standard deviation of 300 mg**.
At $\alpha = 0.01$, test whether the **average sodium intake differs** from the recommended value.
20. A machine fills cereal boxes that should contain **500 g** on average. A sample of **100 boxes** has a **mean weight of 505 g**, and $\sigma = 8$ g.
At $\alpha = 0.05$, test whether the machine is **overfilling** the boxes.
21. A survey found that **60%** of customers prefer brand A. A rival company wants to test if the proportion has **changed**.
In a new sample of **400 customers**, **250** preferred brand A.
At $\alpha = 0.05$, test whether the **true proportion differs from 0.60**.
22. A college claims that at least **70%** of its graduates find jobs within 3 months. A survey of **120 graduates** shows **78** found jobs within that time.
At $\alpha = 0.05$, test whether **fewer than 70%** find jobs (i.e., test if the claim is overstated).
23. A website claims that **90%** of users are satisfied. In a sample of **200 users**, **170** say they are satisfied.
At $\alpha = 0.01$, test whether the **satisfaction rate is below 90%**.
24. A nutrition bar company claims that the **average protein content** of their bars is **20 g**. A sample of **16 bars** shows a **mean of 19.2 g** and a **sample standard deviation of 1.4 g**.
At $\alpha = 0.05$, test whether the **mean protein content differs from 20 g**.

25. A delivery service claims the **average delivery time** is **30 minutes**. A random sample of **64 deliveries** shows a **mean of 32 minutes**, with a **population standard deviation of 6 minutes**.

At $\alpha = 0.05$, test whether the **average delivery time is greater than 30 minutes**.

26. A machine is designed to fill bottles with **500 ml** of juice.

A random sample of **40 bottles** has the following fill amounts (in ml):

498, 503, 501, 499, 502, 504, 497, 500, 503, 499,
501, 505, 496, 500, 502, 499, 504, 498, 503, 500,
497, 505, 502, 498, 504, 499, 501, 500, 497, 503,
499, 502, 500, 503, 497, 505, 504, 498, 501, 500

Population standard deviation: $\sigma = 4$ ml

Tasks:

- Compute sample mean ($\bar{x}$).
- Formulate hypotheses to test if the machine fills correctly ($\mu = 500$).
- Compute standard error, margin of error (95% CI), and interval estimate.
- Perform hypothesis test using **p-value** and **critical value** (Z-test).
- Interpret results.

27. A company claims its energy drink contains **80 mg of caffeine** on average. A random sample of **30 bottles** gives these caffeine contents (mg):

78, 81, 79, 80, 83, 77, 82, 79, 81, 84, 80, 82, 79, 78, 83, 77, 82, 79, 81, 80, 84, 83, 78, 79, 82,
80, 81, 79, 82, 83

$\sigma = 3$ mg

Test at $\alpha = 0.05$ if the mean caffeine content is **greater than 80 mg**.

Also find 95% CI, SE, and ME.

28. A dietitian believes that the average sodium content of a soup is **too high** compared to **650 mg**.

A sample of **10 cans** gives the following sodium contents (mg):

660, 645, 668, 652, 649, 655, 671, 648, 656, 659

Find:

- Sample mean and sample standard deviation.
- SE, ME, and 95% CI.
- Develop hypotheses ($H_0: \mu = 650$).
- Perform **test** using both **p-value** and **t-critical**.

29. A class's average test score is claimed to be **75**.

A teacher thinks the score is **lower**.

A sample of **8 students** scores:

72, 70, 74, 76, 71, 73, 69, 75

Test at $\alpha = 0.05$ using the **t-test**.

Also find SE, ME, and 95% CI.

30. A car manufacturer claims its model accelerates from 0–60 mph in **7.5**

seconds. A sample of **50 cars** gives a mean of **7.3 seconds**, with $\sigma = 0.5$ seconds. •

Test if cars accelerate **faster than claimed** ($\mu < 7.5$).

- Use both **p-value** and **critical Z**.
- Find 99% confidence interval.

31. An instructor believes the **average GPA** of students in her department is

3.0. A random sample of **12 students** gives GPAs:

3.1, 2.8, 3.3, 3.0, 3.2, 2.9, 3.4, 3.1, 2.7, 3.0, 3.2, 3.1

Test if the mean GPA differs from 3.0 ($\alpha = 0.05$).

Compute SE, ME, CI, and perform **t-test**.

32. A manufacturer claims **90%** of its devices pass quality tests.

In a sample of **200 devices**, **174 passed**.

Test at $\alpha = 0.05$ if the proportion differs from 0.90.

Find SE, ME, and 95% CI.

33. A university claims that **60%** of its graduates find jobs within 6 months. In a survey of **120 graduates**, **65 found jobs**.

At $\alpha = 0.01$, test if the job rate is **less than 60%**.

Compute SE, ME, and 99% CI.

34. A marketing campaign claims that at least **40%** of customers prefer a new product. Out of **250 surveyed**, **115 preferred it**.

Test at $\alpha = 0.05$ if the proportion exceeds 0.40.

Also find 95% CI, SE, and ME.

35. An engineer wants to check if a new alloy's **tensile strength** exceeds **200**

MPa. A sample of **7 test pieces** shows strengths (MPa):

198, 205, 202, 203, 199, 206, 204

Test at $\alpha = 0.05$ (right-tailed t-test).

Compute:

- Sample mean, $\bar{x}$
- SE, ME
- 95% CI
- Perform test using both **p-value** and **critical t**

36. A beverage company compares average sugar content (grams) of two recipes A and B. Two independent large production lines were sampled.

• Sample A: ($n_1 = 50$), ($\bar{x}_1 = 12.4$) g. Population SD ($\sigma_1 = 2.0$) g.

• Sample B: ($n_2 = 60$), ($\bar{x}_2 = 11.7$) g. Population SD ($\sigma_2 = 2.5$) g.

Test at ($\alpha = 0.05$) whether the mean sugar contents **differ** (two-tailed). Also produce the

95% confidence interval for ($\mu_1 - \mu_2$).

Tasks:

- Compute the sample difference ($\bar{x}_1 - \bar{x}_2$).
- State (H_0) and (H_a).
- Compute SE for ($\bar{x}_1 - \bar{x}_2$) using known σ s.
- Compute margin of error and 95% CI.
- Find test statistic (Z), p-value, and compare with critical Z.
- State conclusion.

37. A machine manufacturer claims a new process increases average tensile strength. Two independent processes are compared.

- Process Old: ($n_1=40$), ($\bar{x}_1=195$) MPa, ($\sigma_1=6$) MPa.
- Process New: ($n_2=35$), ($\bar{x}_2=199$) MPa, ($\sigma_2=7$) MPa.

Test at ($\alpha=0.01$) whether the new process yields **higher mean strength** ($H_a: \mu_1 > \mu_2$). Provide the **99% CI** for ($\mu_1 - \mu_2$).

Tasks: same checklist (use right-tailed Z test).

38. A nutrition study compares average daily protein intake for men and women in a sample.

- Men: ($n_1=28$), ($\bar{x}_1=75$) g, ($s_1=12$) g.
- Women: ($n_2=32$), ($\bar{x}_2=68$) g, ($s_2=10$) g.

Use ($\alpha=0.05$) to test whether mean protein intake differs between sexes. Use the appropriate **Welch t-test** (unequal variances). Also compute **95% CI** for

($\mu_1 - \mu_2$). **Tasks:**

1. Compute ($\bar{x}_1 - \bar{x}_2$).
2. State hypotheses (two-tailed).
3. Compute SE for difference using (s_1, s_2).
4. Compute degrees of freedom (Welch approximation), t-critical, margin, CI.
5. Compute t statistic, p-value, and compare with critical t.
6. Conclude.

39. A pharmaceutical trial compares mean blood pressure reduction for two drugs. Variances can be assumed equal.

- Drug A: ($n_1=22$), ($\bar{x}_1=12.5$) mmHg, ($s_1=3.1$).
- Drug B: ($n_2=20$), ($\bar{x}_2=10.8$) mmHg, ($s_2=2.9$).

Test at ($\alpha=0.05$) whether the mean reductions differ. Use **pooled t-test** and give a **95% CI** for ($\mu_1 - \mu_2$).

Tasks: compute pooled variance, SE (pooled), t-statistic, df, p-value, critical t, ME, CI.

40. Researchers compare the mean reaction time of two new drug formulations.

- Formulation X: ($n_1=9$), ($\bar{x}_1=1.85$) s, ($s_1=0.12$) s.
- Formulation Y: ($n_2=7$), ($\bar{x}_2=1.77$) s, ($s_2=0.10$) s.

Assume unequal variances. Test at ($\alpha=0.05$) if mean reaction times differ. Provide **95% CI** for ($\mu_1 - \mu_2$).

Tasks: use Welch t-test, compute df, SE, t, p-value, critical t, CI.

41. A company tests two ad designs and records whether viewers clicked (success). • Ad A: ($n_1=800$), clicks ($\bar{x}_1=176$).

• Ad B: ($n_2=750$), clicks ($\bar{x}_2=150$).

Test at ($\alpha=0.05$) whether the click-through proportions differ ($H_0: p_1=p_2$). Also produce the **95% CI** for (p_1-p_2).

Tasks:

1. Compute ($\bar{p}_1, \bar{p}_2$) and pooled ($\bar{p}$).
2. Compute SE (pooled) for hypothesis test and unpooled SE for CI.
3. Compute Z statistic, p-value, critical Z, ME, and CI.
4. Interpret.

42. A hospital claims new follow-up calls reduce readmission proportion below the current 12%. A pilot compares old vs new program.

• Old program: ($n_1=400$), readmissions ($\bar{x}_1=52$).

• New program: ($n_2=420$), readmissions ($\bar{x}_2=36$).

Test at ($\alpha=0.01$) whether the new program has a **lower** readmission proportion ($H_a: p_1 < p_2$). Also find the **99% CI** for (p_1-p_2).

Tasks: pooled test for p difference (left-tailed), p-value and critical Z, plus CI.

43. A supermarket trial tests whether a new display increases the proportion of customers buying an add-on product.

• Control: ($n_1=500$), purchases ($x_1=65$).

• Display: ($n_2=520$), purchases ($x_1=88$).

Test at ($\alpha=0.05$) whether the new display increases the purchase proportion ($H_a: p_1 < p_2$). Also compute **95% CI** for (p_1-p_2).

Tasks: compute ($\bar{p}$), pooled test, Z, p-value (right-tailed), critical Z, ME, CI.

44. A snack company believes customer preferences for its snacks follow these proportions:

Chips	Type Claimed
Cookies	portion 0.40
Nuts	0.30
Popcorn	0.20
	0.10

In a survey of **200** customers, the results are:

Snack Typ	
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Observed	90
Cookies	65
Nuts	30
Popcorn	15

Test at $\alpha = 0.05$ whether customer preferences follow the claimed distribution.

45. A car dealer claims that the color preferences for buying a new car are: **Color Theoretical %**

White 25%

Black 30%

Gray 20%

Blue 15%

Other 10%

A sample of **300** new car buyers shows:

Color Observed

White 65

Black 90

Gray 70

Blue 40

Other 35

Test the goodness-of-fit at $\alpha = 0.01$.

46. Past medical data suggest the population blood type distribution is:

Blood Type %

A 42%

B 10%

AB 4%

O 44%

A random sample of **500** patients reveals:

Blood Type	
Observed	
A	210
B	
AB	48

O	22
	220

Is the sample consistent with the known distribution? Use $\alpha = 0.05$.

47. A university suggests that students prefer studying in:

Location Claimed Proportion

Library 0.50

Dorm 0.35

Coffee Shop 0.15

A survey of **120** students reports:

Location	Observed
Library	
Dorm	66
Coffee Shop	40

Test at $\alpha = 0.10$ whether the actual preferences fit the university claim.

48. An online learning site expects enrollment distribution in course categories as:

Category Expected Proportion

Technology 0.25

Business 0.20

Arts 0.10

Language 0.15

Health 0.20

Other 0.10

From **400** enrollments, the counts are:

Category	Observed
Technology	
Business	
Arts	70
Language	30

Health	55
Other	90
	45

Test the claim using $\alpha = 0.05$.

49. Gender vs. Music Preference

A survey records preferred music genre by gender:

Male			Total	90
Female				110
50 Total	30	42	18	200
		30	30	
	80	72	48	

Test at $\alpha = 0.05$:

Is music preference independent of gender?

50. Smoking Habit vs. Exercise Frequency

Data collected from 300 adults:

Smoker		Regular Exercise		100
Non-Smoker		Occasional Exercise		200
80 Total		No Exercise		300
	20	45	35	
		90	30	
	100	135	65	

Test at $\alpha = 0.01$:

Are smoking and exercise habits independent?

51. Customer Age Group vs. Preferred Shopping Method

A retailer collects data:

Age Group			180
In-Store Total	120	60	15
36–55			

56+ Total	90	60	0
	50	70	12
	260	190	45
			0

Test at $\alpha = 0.05$:

Is preferred shopping method related to age group?

52. Employee Department vs. Job Satisfaction

Survey responses:

Department				50
Neutral Dis				10
TotalHR	25	10	15	0
Sales	40	25	35	70
IT	35	20	15	22
Total	100	55	65	0

Test at $\alpha = 0.05$:

Is job satisfaction independent of department?

53. A poll asks whether voters support, oppose, or are neutral toward a new policy:

Political Party				12
Oppose Neutr				0
TotalParty A	60	35	25	13
Party B	45	55	30	0
Party C	30	25	15	70
Total	135	115	70	32
				0

Test at $\alpha = 0.05$:

Are policy opinions independent of political affiliation?

54. Training Methods

A manager wants to test whether three different employee training programs lead to different productivity scores.

Group			ize Mean Score
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Method		82	Standard Deviation 5
14 Methods		88	
Method B		91	
Method 10			

Test at $\alpha = 0.05$:

Are the mean productivity scores significantly different among training methods? Create ANOVA Table

55. Fertilizer Types

A botanist tests three fertilizers on plant height (cm). Data collected:

- Fertilizer A: 42, 45, 40, 47, 43
- Fertilizer B: 50, 55, 52, 53, 56
- Fertilizer C: 48, 45, 47, 49, 46

Test at $\alpha = 0.01$ if the fertilizers produce different average plant heights. Create ANOVA Table

56. Diet Plans and Weight Loss

A study tests four diet plans measuring weight loss over 6 weeks:

Diet A		Sample Size	Mean	1.1
Diet B			Std. Dev.	1.3
Diet C	20	4.2		1.0
Diet D	18	5.0		1.4
	22	6.1		
	16	4.8		

Use $\alpha = 0.05$:

Are the mean weight losses different across diets?
Create ANOVA Table

57. Student Exam Performance by Teaching Style

A teacher tries 3 teaching approaches:

- Approach 1: 67, 72, 70, 65, 68, 74
- Approach 2: 75, 78, 82, 79, 77
- Approach 3: 60, 62, 58, 65, 61, 59

At $\alpha = 0.05$, determine if teaching style affects exam

performance. Create ANOVA Table

58. Average Wait Times at Restaurants

A researcher compares wait times (minutes) at 4 restaurants:

R1	15, 18, 20, 22, 19, 21
R2	18, 20, 22, 19, 21
R3	25, 28, 30, 27, 26
R4	15, 17, 14, 16, 18
	20, 22, 19, 21, 23

Test at $\alpha = 0.05$:

Do the restaurants have different mean wait times?

Create ANOVA Table

59. Pain Relief Medication

4 medications tested on 5 patients (blocks) using pain score reduction:

Patient				13
Med B				14
DP1	12	14	10	12
P2	11	15	9	15
P3	10	13	11	14
P4	13	16	12	
P5	12	14	10	

Test at $\alpha = 0.05$

Create ANOVA Table

60. Machine Performance

3 lubrication oils tested under 6 machines (blocks) using output rate:

Machine			60
Oil CM1			59
M2	55	58	63
M3	52	57	61
M4	56	60	62
M5	54	59	60

M6	53	56	
	55	58	

Test at $\alpha = 0.01$

Create ANOVA Table

61. Fertilizer Efficiency

4 fertilizers tested in 4 plots, yield (kg):

Plot				
F41				
2				
3				
4				

Test at $\alpha = 0.05$

Create ANOVA Table

62. Training Methods

3 methods tested on 5 instructors for performance rating:

Instructor			
T3I1		7	
I2		8	
I3		8	
I4		2	
I5		8	
		0	
		7	
		5	
		8	
		1	
		7	
		9	
		7	

		7	
		8	
		3	
		8	
		2	
		8	
		0	
		8	
		4	
		8	
		1	
		7	
		6	
		8	
		2	
		8	
		0	

Test at $\alpha = 0.05$

Create ANOVA Table

63. Taste Testing of Coffee

5 coffee blends tested by 4 expert tasters (rating out of 10):

8	7	9	8
7	8	8	7
8	7	9	8

A 7 B 8 C 8 D 7 8 8 7 7

Test at $\alpha = 0.10$

Create ANOVA Table

64. Battery Life

Factors:

A = Brand (A1, A2)

B = Temperature (Low, High)

Replications = 3

			low gh	
18	19	17	14	15

A1 13 A2 20 21 19 17 16 18 Test at α
= 0.05

Create ANOVA Table

65. Crop Growth

A = Fertilizer (F1, F2, F3)

B = Irrigation (I1, I2)

Replications = 2

Fertilizer				
I2F1				
F2				
F3				

Test at $\alpha = 0.01$

Create ANOVA Table

66. Teaching Method $\times$ Exam Level

A = Teaching Method (M1, M2)

B = Exam Level (Basic, Advanced)

Replications = 4

M1		Basic Scores cores 76, 80, 78, 69 82, 84, 85, 83 8, 76, 77
M2		

Test at $\alpha = 0.05$

Create ANOVA Table

67. Packaging × Display

A = Packaging Type (Box, Bottle)

B = Store Display (Front, Side, Back)

Replications = 3

Sales Units (raw data):

Packaging		Side	Back
Front Box		120,122,118	
Bottle		110,112,109	
		105,107,106	
		130,131,128	
		125,124,123	
		110,112,111	

Test at $\alpha = 0.05$

Create ANOVA Table

68. Fuel Type × Engine Design

A = Fuel Type (Gasoline, Ethanol, Hybrid)

B = Engine Design (Design 1, Design 2)

Replications = 3

Fuel		D2 Data
Ethanol	Gasoline 225, 230, 228	
Hybrid	195, 198, 191	
	225, 230, 228	

Test at $\alpha = 0.01$

Create ANOVA Table

69. Quality Inspection

A quality engineer records whether an item produced each hour is Good (G) or

Defective (D):

Sequence (20 items):

G, G, D, G, D, D, G, G, G, D, G, D, D, D, G, G, D, G, D, G

Test ($\alpha = 0.05$): Are the data random?

70. Stock Market Movement

Daily stock price change:

+ = Increase, - = Decrease

Sequence (18 days):

+, -, -, +, +, -, -, -, +, +, +, -, +, -, -, +, +, -

Test randomness at $\alpha = 0.05$.

71. Machine Output

Classify output above or below median (B = Below, A =

Above): Sequence (15 outputs):

B, A, A, B, B, A, B, A, A, A, B, B, A, B, B

Is the sequence random? Use $\alpha = 0.10$.

72. Customer Arrivals

“L” = Low arrivals, “H” = High arrivals (based on threshold): L, H, L, L, H, H, H, L, H, H, L, L, L, H

Test randomness at $\alpha = 0.05$.

73. Website Traffic

Classified relative to average:

H, H, H, L, L, H, L, H, H, L, L, L, H, H, L, H

Test at $\alpha = 0.05$ whether highs and lows are random.

74. Diet Effect

Weight before vs after 8-week diet (kg):

Person	Before	After
1	80	73
2	82	89
3	75	83
4	90	75
5	85	90
6	88	88
7	78	79
8	92	
	88	

	81	
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Test: Does the diet reduce weight? ($\alpha = 0.05$, one-tailed)

75. Teaching Method Effect

Test scores before vs after new teaching method:

Differences (After – Before):

3, 5, 4, –1, 2, –2, 3, 1, 4, –1

Test at $\alpha = 0.05$ whether method improves scores (one-tailed).

76. Pain Reduction

Pain scores (lower is better):

Before: 7, 8, 6, 7, 9, 8

After: 5, 7, 6, 6, 7, 8

At $\alpha = 0.10$, test whether treatment reduces pain (one-tailed).

77. Median Comparison

Service response times vs a claimed median of 15 minutes:

Sample data:

16, 14, 15, 13, 18, 17, 14, 16, 12, 19

Test ($\alpha = 0.05$, two-tailed) if the median is different from 15.

78. Machine Calibration

Paired measurements: Standard vs Machine output

Difference (Machine – Standard):

1, 0, –2, –1, 1, 2, 0, –1, –2, 1

Test ($\alpha = 0.05$, two-tailed): Does the machine produce a median difference of 0?

79. WSR-1: Meditation & Stress

A researcher measures stress scores before and after a 2-week meditation program:

| Before | 18 | 21 | 23 | 20 | 19 | 24 | 22 | 25 |

| After | 16 | 18 | 21 | 19 | 17 | 22 | 21 | 23 |

Test at $\alpha = 0.05$: Does meditation reduce stress?

80. WSR-2: Training Program for Typing Speed

Typing speed (wpm) before & after training:

| Before | 52 | 55 | 60 | 57 | 58 | 54 | 59 |

| After | 56 | 58 | 63 | 60 | 61 | 57 | 62 |

Test if training improves typing speed (one-tailed, $\alpha = 0.01$)

81. WSR-3: Pain Treatment on Joint Pain

Pain scale: higher = more pain

| Before | 9 | 7 | 8 | 10 | 9 | 6 |

| After | 6 | 6 | 7 | 8 | 7 | 5 |

Test at $\alpha = 0.05$ whether treatment reduces pain.

82. WSR-4: Productivity Before & After Workplace Lighting Change | Before | 42 | 45 | 44 | 46 | 47 | 43 | 48 | 44 |
| After | 44 | 46 | 46 | 47 | 49 | 45 | 50 | 47 |
Test if new lighting improves productivity (two-tailed)

83. WSR-5: Memory Test Using Flashcards
Scores:
Before: 72, 70, 68, 75, 77, 73, 69
After: 75, 72, 70, 77, 79, 75, 71
Test at $\alpha = 0.05$ for improvement.

84. MW-1: Smartphone Battery Life
Two brands tested (hours):
Brand A: 10, 12, 9, 11, 13, 10
Brand B: 14, 13, 15, 12, 16, 14
Test at $\alpha = 0.05$ if mean life differs.

85. MW-2: Customer Satisfaction by Two Stores
Rating (1–10):
Store A: 8, 7, 9, 6, 8
Store B: 6, 5, 7, 6, 7
One-tailed test at $\alpha = 0.05$: Store A higher satisfaction?

86. MW-3: Study Strategy Comparison
Test scores:
Strategy A: 75, 78, 80, 74, 76
Strategy B: 82, 85, 81, 83, 84
Test $\alpha = 0.01$, any performance difference?

87. MW-4: Reaction Times for Two Game Controllers
Time (ms):
Controller X: 220, 215, 230, 225, 218
Controller Y: 205, 200, 210, 208, 207
At $\alpha = 0.05$, is controller Y faster?

88. MW-5: Sales Training Approaches
Sales achieved (units):
Training A: 52, 48, 54, 55, 50
Training B: 56, 53, 59, 57, 54
Test for difference in effectiveness at $\alpha = 0.05$.

89. KW-1: Employee Engagement by Department
Engagement scores:
HR: 72, 75, 71, 73
Sales: 78, 80, 79, 77
IT: 70, 68, 69, 71

Test at $\alpha = 0.05$: Same medians?

90. KW-2: Crop Yield under 3 Fertilizers

Yield:

F1: 20, 22, 21, 23

F2: 25, 24, 26, 27

F3: 18, 17, 19, 18

Test at $\alpha = 0.01$.

91. KW-3: Delivery Time for Different Courier

Services Delivery (days):

C1: 3, 4, 3, 2, 4

C2: 5, 6, 5, 4, 5

C3: 4, 4, 3, 5, 4

Test $\alpha = 0.05$.

92. KW-4: Customer Ratings for 4 App

Versions Ratings (out of 10):

V1: 7, 6, 7, 8

V2: 8, 9, 7, 8

V3: 6, 5, 7, 6

V4: 8, 7, 8, 9

Test for difference at $\alpha = 0.05$.

93. KW-5: Sleep Hours Across 3 Cities

Hours slept nightly:

City A: 7, 6, 8, 7, 7

City B: 6, 5, 6, 5, 6

City C: 8, 7, 8, 9, 7

Test $\alpha = 0.05$.

94. RC-1: IQ vs Academic Score

10 students ranked in IQ and exam score:

IQ Rank: 3, 1, 4, 2, 5, 8, 6, 9, 7, 10

Exam Rank: 2, 1, 3, 4, 5, 7, 6, 8, 9, 10

Test correlation at $\alpha = 0.05$.

95. RC-2: Social Media Followers vs Content

Quality 12 influencers:

Followers Rank: 1-12

Quality Rank: 3,1,4,2,5,7,6,9,8,11,10,12

Test positive correlation ($\alpha = 0.01$, one-tailed)

96. RC-3: Sports Performance vs Motivation

Score 10 athletes ranked by performance and

motivation Performance: 1–10

Motivation: 2,4,1,3,5,6,8,7,9,10

Test $\alpha = 0.10$.